appear to be an objective fact in the world, something capable of creating interference patterns plain for all to see?

Some physicists tried to address these questions about information-theoretic interpretations by stating that the wave function was information about an unseen world underlying quantum physics, one that obeyed different and as yet undiscovered laws. But such a world would have to be nonlocal to satisfy Bell's theorem—in which case, much of the appeal of the information-theoretic interpretations was lost. (Wheeler himself misunderstood the Bell experiments as ruling out determinism, rather than locality.) Others attempted to get around Bell's theorem by altering the laws of probability or breaking one of the other handful of assumptions that entered into Bell's proof—but each of these solutions came with its own strange and difficult problems.

None of these problems meant that information-theoretic interpretations couldn't work. These were challenges that would have to be met or convincingly dismissed, and physicists and philosophers interested in information-theoretic interpretations worked on doing just that. But, for some physicists, the simple idea of the wave function as "information" held the same allure as decoherence: the promise of a quick and easy way to dismiss niggling doubts about the measurement problem. Wheeler said that his inspiration for "it from bit" was Bohr's approach to quantum physics; some took this to mean that this was what Bohr himself had meant all along, that the Copenhagen interpretation was and always had been merely stating that the wave function was information (flatly refusing to answer what it was information about), that this was the One True Way to "understand" quantum physics.

Bell, of course, knew that there was nothing in quantum physics or in his own theorems that led inexorably to the Copenhagen interpretation. He had been promoting pilot-wave theory for decades to illustrate precisely this point. "Why is the pilot wave picture ignored in text books?" Bell asked in 1982. "Should it not be taught, not as the only way, but as an antidote to the prevailing complacency? To show that vagueness, subjectivity, and